

Name:

Date:

Period:

Chapters 16-21: Electricity

Homework Check A (collected Mon, Apr 28)

Chapter 16

Coulomb's Law p. 469 #1, 2, 3, 6, 7, 11 Complete by Mon, Apr 21

STAMP
HERE
5 POINTS

Electric Field pp. 469-470 #19, 20, 22, 25, 32 Complete by Fri, Apr 25

STAMP
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5 POINTS

Ch 16 Conceptual Questions p. 220 #2, 8, 10, 14, 15 Complete by Fri, Apr 25

THESE QUESTIONS SHOULD HAVE AT LEAST ONE FULL SENTENCE OF EXPLANATION

STAMP
HERE
4 POINTS

Chapter 17

Electric Potential p. 496 #1, 2, 9 Complete by Mon, Apr 28

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3 POINTS

Ch 17 Conceptual Questions p. 494 #1, 2, 6 Complete by Mon, Apr 28

THESE QUESTIONS SHOULD HAVE AT LEAST ONE FULL SENTENCE OF EXPLANATION

STAMP
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3 POINTS

Answers

Ch 16

1. $2.7 \times 10^{-3} \text{ N}$
2. 3×10^{14} electrons
3. $2.2 \times 10^4 \text{ N}$
6. 2.69 N
7. 3.76 cm

11. 115.8 N, left
564.2 N, right
448.4 N, left
19. $3.94 \times 10^{-16} \text{ N}$, due west
20. $1.16 \times 10^5 \text{ N/C}$, south
22. 8.8×10^5 , upward
25. (drawing)

32. 65 cm

Ch 17

1. $5.0 \times 10^{-4} \text{ J}$
2. $2.72 \times 10^{-17} \text{ J}$
9. 42.5 kV

Name:

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Equations

Electrostatics Equations

$$F = \frac{kq_1q_2}{r^2}$$

$$E \equiv \frac{F}{q} = \frac{kq}{r^2}$$

$$\Delta V \equiv \frac{\Delta PE}{q} = -\frac{W}{q}$$

Constants

Coulomb Constant	$k = 9.0 \times 10^9 \text{ Nm/C}^2$
Charge of electron/proton	$e = \pm 1.60 \times 10^{-19} \text{ C}$
Electron mass	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Proton mass	$m_p = 1.673 \times 10^{-27} \text{ kg}$
Neutron mass	$m_n = 1.675 \times 10^{-27} \text{ kg}$

Name:

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Chapters 16-21: Electricity

Homework Check B (collected on Test Day)

Chapter 18

Current, Resistance, Ohm's Law p. 521 #1, 2, 4, 7 Complete by Wed, Apr 30

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3 POINTS

Ch 18 Conceptual Questions p. 520 #2, 3 Complete by Wed, Apr 30
THESE QUESTIONS SHOULD HAVE AT LEAST ONE FULL SENTENCE OF EXPLANATION

STAMP
HERE
2 POINTS

Chapter 19

Resistors in Series and Parallel p. 552 #4, 5, 6, 16 Complete by Mon, May 5

STAMP
HERE
3 POINTS

Equivalent Resistance Worksheet Complete by Mon, May 5

STAMP
HERE
4 POINTS

Ch 19 Conceptual Questions p. 549 #4, 11, 12 Complete by Mon, May 5
THESE QUESTIONS SHOULD HAVE AT LEAST ONE FULL SENTENCE OF EXPLANATION

STAMP
HERE
3 POINTS

Bonus Problems! Ch 16 #15 & 31; Ch 19 #19 Turn in separately on test day!

Test will be on Tue, May 6

Name:

Date:

Period:

Answers

Ch 18

1. 10^{19} e/sec
2. 33.5 A-hr or 1.2×10^5 C
4. 26.1 ohms
7. 27.9 A; 8.4×10^4 C

5. 330 ohms; 8.86 ohms
6. Max resistatnce: 2570 ohms; Min resistance: 59.4 ohms
- 16.

Ch 19

4. 8.81 V

- Equivalent: $R = 1346.6$ ohm; $I = 8.9$ mA
- 990 ohm: $V = 8.8$ V; $I = 8.9$ mA
- 680 ohm: $V = 3.2$ V; $I = 4.3$ mA
- 750 ohm: $V = 3.2$ V; $I = 4.7$ mA

Equations

Electrostatics Equations

$$F = \frac{kq_1q_2}{r^2}$$

$$E \equiv \frac{F}{q} = \frac{kq}{r^2}$$

$$V \equiv \frac{PE}{q} = -\frac{W}{q}$$

Circuit Equations

$$I \equiv \frac{\Delta q}{t}$$

$$V = IR$$

$$P = IV = I^2R = \frac{V^2}{R}$$

$$R_{eq} = R_1 + R_2 + \dots$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

Constants

Coulomb Constant	$k = 9.0 \times 10^9 \text{ Nm/C}^2$
Charge of electron/proton	$e = \pm 1.60 \times 10^{-19} \text{ C}$
Electron mass	$m_e = 9.11 \times 10^{-31} \text{ kg}$
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